Exploring How True Information Can Rival Disinformation in Going Viral – Milestone 2

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Falsehoods often seem to spread “farther, faster, deeper, and more broadly” on social media than factual content (Vosoughi, Roy, & Aral, 2018). Sensational disinformation can capture attention and disseminate across the online town square remarkably fast, frequently extinguishing any attempt at a response. This analysis aims to identify strategies and mechanisms to help accurate, true information achieve virality comparable to or greater than that of disinformation. In other words, how can we make the truth as compelling and widely shared as the lies?

Research has demonstrated that false news spreads faster than true news on social platforms like Twitter. A study by Vosoughi et al. (2018) found that false news was 70% more likely to be retweeted than true news and reached 1,500 people six times faster on average. This indicates that lies have a built-in advantage in terms of speed and reach.

There has been plenty of research exploring how to stem the flow of disinformation. MIT Sloan published a study showing that asking social media users to consider the accuracy of the news they are about to share leads to improved content quality (MIT Sloan Office of Communications, 2021). A later study by Altay et al. (2025) found that “prompting social media users to follow news organizations increases… awareness of true news stories.” And recently, Slaughter et al. (2025) noted that “crowd-sourced fact-checking” via the Community Notes feature on X “reduces the depth and structural virality” of all noted posts, regardless of if true or false.

However, I believe that simply diminishing its distribution is not enough. Stopping false information from spreading is only half the battle: we must also overwhelm the town square with true information. False stories enjoy virality because humans are habitually

driven to share sensational or provocative claims that trigger strong emotions (Ceylan, Anderson, & Wood, 2023). If our aim is to preserve and disseminate truth, then we need strategies that drive humans to share true stories with at least the same earnestness as false stories.

Metrics Of a Successful Study

A successful study will present the general elements of factual posts that either successfully or unsuccessfully regularly attain high engagement metrics (views, shares, likes, and comments) on par with or exceeding those of trending false narratives. Ideally, we would develop a formula that would reliably increase engagement metrics on truthful posts, but evidence of practices that do not work would be useful as well. Our hope is to identify statistically significant social media practices that consistently see true information being reshared by users at rates similar to popular disinformation and traveling as fast through networks.

If we can crack the code to boost truth virality, the benefits would be widespread. Social media users and society at large would receive more reliable information, reducing harm from rumors and lies. Public agencies, health experts, and educators would be better able to broadcast life-saving facts at scale. Journalists and fact-checkers would gain confidence that their hard work clarifying facts won't be drowned out by falsehoods. Businesses and brands stand to gain, as their communications and reputations would be less likely to be undermined by viral disinformation campaigns. Social media platform companies themselves would enjoy greater trust and user satisfaction if truthful content thrives, potentially easing regulatory pressures. By understanding the dynamics of

information spread and focusing on making truth as engaging and shareable as lies, we can create a more informed and resilient society.

Defining Data Requirements

To answer the question, "How can true information rival disinformation in going viral?", we need to analyze the spread and engagement of both true and false content on social media.

Needed Variables

To effectively analyze the spread of true versus false information on social media, several key data features must be considered.

Posts can be categorized by content type, labeling posts as true, false, corrections, debunks, opinions, or satire. These posts can be further categorized by visual elements, such as images, memes, videos, and infographics. Sentiment scores, such as anger, awe, surprise, sadness, trust, or disgust, capture the emotional response elicited by a post, while novelty scores indicate whether the content presents new or unexpected information (Trunfio & Rossi, 2021). These classifications provide a foundation for understanding nature and intent behind each post.

Engagement metrics help measure virality by counting shares, likes, comments, and impressions. Number of followers, network centrality, and community membership can reveal how information travels through social networks and the influence of specific users or groups (Huynh, et al., 2022). Temporal data, including timestamps of posting and engagement events, track when and how quickly content gains traction.

Source credibility is crucial in determining the trustworthiness of information. Data on account type (influencer, expert, news outlet, regular user) and verification status help assess the reliability of those disseminating content. Algorithmic signals, such as whether a post is boosted, labeled, or suppressed by platform algorithms, inform how the platform itself influences the spread of information and can significantly affect engagement.

Exposure to prebunking or media literacy interventions should also be considered. Data on whether users have interacted with educational content about misinformation can shed light on the effectiveness of such measures in promoting the dissemination of true information over falsehoods.

Data for this research can be sourced from public social media datasets (e.g., Twitter/X API, Facebook CrowdTangle, Reddit Pushshift, YouTube Data API), fact-checking organizations (Full Fact, Snopes, PolitiFact, Community Notes), and academic resources like MIT's "False News" dataset, PHEME, and FakeNewsNet. Surveys and experiments can provide insights into user sharing habits and responses to prebunking. Platform transparency reports may offer data on algorithms and engagement. For unavailable data, custom methods such as permitted web scraping or collaborating with platforms for anonymized data can be used.

Data Preparation Plan

For robust analysis, datasets from multiple platforms will be integrated, variables and timestamps standardized (UNIX), and engagement metrics normalized. Fact-checking databases will label posts as true or false, while natural language processing (NLP) models

will assess sentiment and content novelty for credibility and virality insights (Hasan, Maisha, & Arifuzzaman, 2019).

Missing engagement metrics will be imputed when possible; posts with essential missing data will be excluded. The dataset will be cleaned of duplicates, bots, and spam, and filtered by relevant periods and topics. Data sources, including behavioral and survey data, will be merged to offer a comprehensive perspective and linked to fact-checking interventions like Community Notes.

All data handling will follow ethical standards, ensure user anonymity, and comply with platform policies and privacy laws.

Data Analysis Plan

Summary statistics will be calculated to determine the averages, medians, and distributions of engagement metrics, with a focus on distinguishing between true and false posts. Following the example of Sangiorgio et al. (2025), we can illustrate virality trends using time series plots, and network graphs to map patterns of information spread, while sentiment heatmaps and bar charts will compare engagement across content types.

Correlation analyses will examine relationships between emotional tone and virality, as well as between source credibility and engagement metrics (Trunfio & Rossi, 2021). Comparative analyses will look at differences between posts that receive algorithmic boosts or fact-check labels and those that do not. The effect of influencer amplification will be contrasted with grassroots sharing (Huynh, et al., 2022). Topic modeling will help identify recurring themes in viral content, distinguishing between true and false information.

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